INCH-POUND

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SUPERSEDING
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PERFORMANCE SPECIFICATION

ACOUSTICAL TRANSMISSION LOSS BARRIER MATERIALS

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of acoustical transmission loss barrier material (see 6.1).

1.2 Classification. Material furnished under this specification is one of the types and classes in table I as specified (see 6.2). Refer to 6.10 for a cross-reference of old to new type designations and substitutability data.

TABLE I. Barrier material.

Type	Description	Class	Nominal maximum surface density (lb/ft^2)
I	Faced, lead-free, low-temperature barrier material	1	0.75
		2	1.0
		3	1.5
II	Faced, lead-free, high-temperature barrier material	2	1.0
		3	1.5
III	Noncombustible barrier material	-	1.0

1.3 Part or identifying number (PIN). The PINs to be used for acoustical transmission loss barrier materials acquired to this specification are created as follows:

M	**24699**	-	**X**	**X**
Prefix for Military Specification	Specification Number	-	Type (see code below)	Class (see code below)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5640

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PIN Codes				
Type	**Code**		**Class**	**Code**
I	A		1	1
II	B		2	2
III	C		3	3

Example PIN: M24699-A2 is faced, lead-free, low-temperature acoustical transmission loss barrier material (type I) with a nominal surface density of 1.0 pound (lb) per square foot (ft) (class 2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32585 - Insulation, Thermal and Acoustic, Fibrous Glass

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

CODE OF FEDERAL REGULATIONS (CFR)

46 CFR 164.009 - Noncombustible Materials for Merchant Vessels

(Copies of this document are available online at www.ecfr.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9510-AB-ATM-010 - Nuclear Powered Submarine Atmosphere Control Manual; Volume 1

(Copies of the chapter titled “Material Control Program” are available by email request to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil.)

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D638	-	Standard Test Method for Tensile Properties of Plastics
ASTM D1388	-	Standard Test Methods for Stiffness of Fabrics
ASTM D2261	-	Standard Test Method for Tearing Strength of Fabrics by the Tongue (Single Rip) Procedure (Constant-Rate-of-Extension Tensile Testing Machine)
ASTM D2724	-	Standard Test Method for Bond Strength of Bonded, Fused, and Laminated Apparel Fabrics
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D5035	-	Standard Test Method for Breaking Force and Elongation of Textile Fabrics (Strip Method)
ASTM E90	-	Standard Test Method for Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions and Elements
ASTM E162	-	Standard Test Method for Surface Flammability of Materials Using a Radiant Heat Energy Source
ASTM E662	-	Standard Test Method for Specific Optical Density of Smoke Generated by Solid Materials
ASTM E800	-	Standard Guide for Measurement of Gases Present or Generated During Fires
ASTM E1123	-	Standard Practices for Mounting Test Specimens for Sound Transmission Loss Testing of Naval and Marine Ship Bulkhead Treatment Materials

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials shall be as specified in 3.2.1 through 3.2.5.

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3.2.1 Toxicity and prohibited materials.

3.2.1.1 Toxicity. When evaluated in accordance with 4.4.1.1, the barrier material shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.4.1.1, 6.2, and 6.5).

3.2.1.2 Prohibited materials. The barrier material shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs (see 6.2).

3.2.1.2.1 Asbestos. Barrier materials shall be fabricated without the use of any type of asbestos.

3.2.2 Type I. Type I barrier material shall consist of a flexible layer with a woven outer facing. Type I barrier material shall be fabricated without the use of lead. Type I barrier material shall be fabricated for use in applications up to 180 °F.

3.2.3 Type II. Type II barrier material shall consist of a flexible layer with a woven outer facing on one side or both sides, as specified (see 6.2). Type II barrier material shall be fabricated without the use of lead. Type II barrier material shall be fabricated for use in applications up to 482 °F.

3.2.4 Type III. Type III barrier material shall consist of a flexible, noncombustible layer.

3.2.5 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Dimensions and quantities.

3.3.1 Width of rolls (type I and type II). Type I and type II barrier material shall be fabricated in rolls. The width of the rolls shall be 26, 38, or 48 inches, or as specified (see 6.2).

3.3.2 Quantity of pieces per roll (type I and type II). The quantity of pieces per roll shall not exceed the values in table II.

TABLE II. Quantity of pieces per roll.

Type	Class	Maximum quantity of pieces per roll
I	1	3
I	2	3
I	3	2
II	2	3
II	3	2

3.3.3 Size of panels (type III). Type III barrier material shall be fabricated in panels that shall be 3 by 4 feet or 3 by 8 feet, or as specified (see 6.2).

3.3.4 Tolerance. The tolerance for dimensions of type I, type II, and type III barrier material shall not exceed ±¼ inch.

3.4 Surface density and weight.

3.4.1 Surface density. Nominal values for surface density shall not exceed the values shown in table III. The surface density of individual samples may vary within a tolerance of ±10 percent of the nominal density.

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TABLE III. Nominal surface density.

Nominal surface density (lb/ft^2)					
Type I, class 1	**Type I, class 2**	**Type I, class 3**	**Type II, class 2**	**Type II, class 3**	**Type III**
0.75	1.0	1.5	1.0	1.5	1.0

3.4.2 Weight of rolls (type I and type II). The weight per roll shall be 100 pounds ±10 percent.

3.5 Breaking force. Breaking force shall be as specified in table IV.

TABLE IV. Minimum breaking force.

Type	Warp breaking force (lb/inch)	Fill breaking force (lb/inch)
I	400	300
II	300	200
III	400	300

3.6 Tearing strength (type I and type II). Minimum tearing strength shall be as specified in table V.

TABLE V. Tearing strength.

Type	Warp tear strength (lb)	Fill tear strength (lb)
I	50	40
II	40	30

3.6.1 Rivet and grommet load bearing strength. Load bearing strength shall be not less than the values specified in table VI.

TABLE VI. Rivet and grommet load bearing strength.

Type	Pounds of pull
I	180
II	100
III	180

3.7 Flexural rigidity. Flexural rigidity values shall not exceed the values in table VII. In addition, the barrier material shall show no visible rupture or cracking on the outer or inner surfaces during and following the test.

TABLE VII. Maximum flexural rigidity.

°F	Inch-pounds
45	1.0
133	0.3

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3.8 Peel strength (type I and type II). Woven outer facing 180-degree peel strength shall be in accordance with table VIII.

TABLE VIII. Minimum woven outer facing peel strength.

Type	Warp (lb/inch)	Fill (lb/inch)
I	4	2
II	4	2

3.9 Fire resistance. Fire resistance shall be as specified below.

3.9.1 Flammability (type I and type II). Materials shall provide a flame spread not to exceed the limits in table IX.

TABLE IX. Flame spread.

Type	Maximum rating
I	30
II	5

3.9.2 Smoke density (type I and type II). Materials shall not exceed the smoke density limits specified in table X.

TABLE X. Smoke density.

Type	Class	Rating
I	1	300
I	2	350
I	3	450
II	2, 3	25

3.9.3 Noncombustibility (type III). Type III barrier material shall be noncombustible.

3.10 Sound transmission loss. Sound transmission loss shall be as specified below.

3.10.1 Type I material. Sound transmission loss values of type I barrier material in decibels (dB) shall be not less than the minimum values specified in tables XI, XII, and XIII.

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TABLE XI. Minimum sound transmission loss values in dB for type I, class 1.

One-third octave band center frequency in hertz (Hz)																		
Hz	100	125	160	200	250	315	400	500	630	800	1000	1250	1600	2000	2500	3150	4000	5000
dB	13	10	10	13	14	15	17	19	21	22	25	26	28	30	32	33	35	37

TABLE XII. Minimum sound transmission loss values in dB for type I, class 2.

One-third octave band center frequency in Hz																		
Hz	100	125	160	200	250	315	400	500	630	800	1000	1250	1600	2000	2500	3150	4000	5000
dB	12	12	12	14	16	17	19	21	23	24	26	28	30	32	34	36	38	39

TABLE XIII. Minimum sound transmission loss values in dB for type I, class 3.

One-third octave band center frequency in Hz																		
Hz	100	125	160	200	250	315	400	500	630	800	1000	1250	1600	2000	2500	3150	4000	5000
dB	15	14	14	18	19	20	22	24	27	28	31	32	33	36	38	40	42	43

3.10.2 Type II material. Sound transmission loss values of type II barrier material in decibels shall be not less than the minimum values specified in tables XIV and XV.

TABLE XIV. Minimum sound transmission loss values in dB for type II, class 2.

One-third octave band center frequency in Hz																		
Hz	100	125	160	200	250	315	400	500	630	800	1000	1250	1600	2000	2500	3150	4000	5000
dB	16	17	16	17	18	19	20	22	24	26	27	29	31	32	34	35	37	39

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TABLE XV. Minimum sound transmission loss values in dB for type II, class 3.

One-third octave band center frequency in Hz																		
Hz	100	125	160	200	250	315	400	500	630	800	1000	1250	1600	2000	2500	3150	4000	5000
dB	19	20	19	21	20	21	22	23	25	28	30	32	34	35	37	39	40	42

3.10.3 Type III material. Sound transmission loss values of type III barrier material in decibels shall be not less than the minimum values specified in table XVI.

TABLE XVI. Minimum sound transmission loss values in dB for type III.

One-third octave band center frequency in Hz																		
Hz	100	125	160	200	250	315	400	500	630	800	1000	1250	1600	2000	2500	3150	4000	5000
dB	15	12	18	28	34	39	43	45	50	53	56	57	57	58	60	65	67	69

3.11 Workmanship. The material shall be free of cracks, scratches, dents, imbedded particles, broken edges or corners, and other defects that could affect serviceability.

3.12 Off-gassing. The barrier material shall be evaluated for off-gassing in accordance with the requirements of 4.4.10. Based on the circumstances of use and the chemical nature of the barrier material, the Navy will determine whether off-gas testing is required or if an administrative assessment is acceptable. In order to be considered acceptable for use in submarines, the barrier material shall be assigned to either the “Permitted” or “Limited” category (see 4.4.10 and 6.6).

3.13 Fire gas toxicity. The barrier material shall be evaluated for types of gases and the toxicity levels released in accordance with 4.4.11.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.1.1 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified herein.

4.2 First article inspection. First article inspection shall be performed on one roll or panel of barrier material when a first article sample is required (see 3.1 and 6.2). First article inspection shall consist of the tests specified in 4.4 and examinations specified in 4.5 as outlined in table XVII.

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TABLE XVII. First article inspection.

Inspection	Requirement paragraph	Test paragraph
Toxicity	3.2.1.1	4.4.1.1
Prohibited materials	3.2.1.2	4.4.1.2
Materials (all types)	3.2.2, 3.2.3, & 3.2.4	4.5.2
Dimensions and quantities:		
Width of rolls (type I and type II)	3.3.1 & 3.3.4	4.4.2
Quantity of pieces per roll (type I and type II)	3.3.2 & 3.3.4	4.5.2
Size of panels (type III)	3.3.3 & 3.3.4	4.4.2
Surface density and weight:		
Surface density	3.4.1	4.4.3.1
Weight of rolls (type I and type II)	3.4.2	4.4.3.2
Breaking force	3.5	4.4.4
Tearing strength (type I and type II)	3.6	4.4.5
Rivet and grommet load bearing strength	3.6.1	4.4.5.1
Flexural rigidity	3.7	4.4.6
Peel strength (type I and type II)	3.8	4.4.7
Fire resistance:		
Flammability (type I and type II)	3.9.1	4.4.8.2
Smoke density (type I and type II)	3.9.2	4.4.8.3
Noncombustibility (type III)	3.9.3	4.4.8.4
Sound transmission loss:		
Type I and type II material	3.10.1 & 3.10.2	4.4.9.1
Type III material	3.10.3	4.4.9.2
Workmanship	3.11	4.5.1
Off-gassing	3.12	4.4.10
Fire gas toxicity	3.13	4.4.11

4.2.1 Guidelines for first article inspection. The tests specified in 4.4.8, 4.4.9, and 4.4.1.1 shall only be conducted for one of the following reasons (see 6.2):

a. If within a 3-year period preceding the date of the invitation for bids, the material has not been tested and found in conformance to 3.9, 3.10, and 3.2.1.1, respectively, or

b. If the material offered for delivery is not manufactured the same in all respects as that previously tested.

4.3 Conformance inspection. Conformance inspection shall be in accordance with table XVIII and 4.3.2.

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TABLE XVIII. Conformance inspection.

Inspections	Requirement paragraph	Test paragraph
Group A		
Materials (all types)	3.2.2, 3.2.3, & 3.2.4	4.5.2
Dimensions and quantities:		
Width of rolls (type I and type II)	3.3.1	4.4.2
Quantity of pieces per roll (type I and type II)	3.3.2	4.5.2
Size of panels (type III)	3.3.3	4.4.2
Surface density and weight:		
Surface density	3.4.1	4.4.3.1
Weight of rolls (type I and type II)	3.4.2	4.4.3.2
Workmanship	3.11	4.5.1
Group B		
Breaking force	3.5	4.4.4
Tearing strength (type I and type II)	3.6	4.4.5
Rivet and grommet load bearing strength	3.6.1	4.4.5.1
Flexural rigidity	3.7	4.4.6
Peel strength (type I and type II)	3.8	4.4.7

4.3.1 Lot. A lot shall consist of all finished material of one size produced in a continuous run (or at the same time and under the same conditions) and offered for delivery at one time. The sampling unit shall be one roll or panel.

4.3.2 Sampling.

4.3.2.1 Sampling for group A conformance inspection. For the group A conformance inspection (see table XVIII), a random sample of barrier material shall be selected from each lot in accordance with ASQ/ANSI Z1.4 at inspection level I. Any defects as specified in table XIX detected during the inspection shall be cause for rejection.

TABLE XIX. Classification of defects.

Examine	Defect	Major	Minor
End Item	Length and width not as specified.		X
	Cracks, scratches, dents, imbedded particles.	X	
	Broken edges or corners (type III).	X	
	Damage or defects affecting function or serviceability.	X	
	Excessive quantity of pieces per roll.	X	

4.3.2.2 Sampling for group B conformance inspection. For the group B conformance inspection (see table XVIII), a random sample of barrier material shall be selected from each lot in accordance with ASQ/ANSI Z1.4 at inspection level S-2. Any failures detected during the inspection shall be cause for rejection.

4.3.2.3 Nonconformance. Instructions regarding failure of group B inspection are identified in 6.9.

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4.4 Tests.

4.4.1 Toxicity and prohibited materials.

4.4.1.1 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.2.1.1 as specified (see 6.2). The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the barrier material using data provided by the manufacturer/distributor to the NMCFHPC (see 3.2.1.1 and 6.5).

4.4.1.2 Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.1.2 and 6.2).

4.4.2 Dimensions. Width of rolls and size of panels shall be determined by using a steel rule calibrated in 1/16-inch graduations (see 3.3.1 and 3.3.3).

4.4.3 Surface density and weight.

4.4.3.1 Surface density. Surface density shall be determined in accordance with ASTM D3776/D3776M (see 3.4.1).

4.4.3.2 Weight of rolls. Weight shall be determined by using a scale with a capacity and sensitivity sufficient to weigh the entire item. The accuracy of the scale shall be certified by a recognized authority (see 3.4.2).

4.4.4 Breaking force. Breaking force shall be determined in accordance with ASTM D5035 (see 3.5).

4.4.5 Tearing strength. Tearing strength shall be determined in accordance with ASTM D2261 (see 3.6).

4.4.5.1 Rivet and grommet load bearing strength. Strength shall be determined using test procedures in accordance with ASTM D638. Test specimens shall be prepared as follows: Test specimens shall be 3 inches wide and 7 inches long. Brass grommets (½-inch inside diameter by 1-inch outside diameter) shall be placed on the width center, 1½ inches from material ends. The two grommeted ends shall be attached to steel hooks mounted in the tension tester jaws so that the steel hooks pass through the grommet eyelets. The tension tester shall be driven at a rate of 20 inches per minute, and the tear-out force of the grommets in pounds of pull shall be recorded (see 3.6.1).

4.4.6 Flexural rigidity. Flexural rigidity shall be determined in accordance with ASTM D1388, option A, Cantilever Test, at 45 °F and 133 °F (see 3.7).

4.4.7 Peel strength. Peel strength shall be determined in accordance with ASTM D2724 (see 3.8).

4.4.8 Fire resistance.

4.4.8.1 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is acceptable to the Technical Authority (see 6.11). Accreditation shall be obtained from a recognized accreditation body such as the American Association for Laboratory Accreditation (A2LA) (www.A2LA.org) or the International Code Council's International Accreditation Services (IAS) (www.IASonline.org). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.7).

4.4.8.2 Flammability. Flammability shall be determined in accordance with ASTM E162 (see 3.9.1) with the barrier material sample positioned so the woven facing faces the heat source.

4.4.8.3 Smoke density. Smoke density shall be determined in accordance with ASTM E662 (see 3.9.2).

4.4.8.4 Noncombustibility. Type III materials shall be tested for noncombustibility in accordance with 46 CFR 164.009 (see 3.9.3).

4.4.9 Sound transmission loss.

4.4.9.1 Type I and type II material. Transmission loss for type I and type II barrier material shall be tested in accordance with ASTM E90 (see 3.10.1).

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4.4.9.2 Type III material. Type III barrier material shall be tested in accordance with ASTM E90, except as follows: The type III barrier material shall be mounted to a base layer of 2-inch thick, fibrous glass panel in accordance with MIL-DTL-32585, type II, form 2, facing A with a nominal density of 3.00 lb/ft^3 (48 kilograms per cubic meter) and installed on a bulkhead in accordance with ASTM E1123 (see 3.10.2). Installation of the type III barrier material shall be in accordance with ASTM E1123, except for the stud installation pattern. The studs shall be installed such that they are aligned in rows and columns in a rectangular pattern, 12 inches apart and 6 inches from the seams in the barrier material.

4.4.10 Off-gassing. The barrier material shall be evaluated for off-gassing in accordance with S9510-AB-ATM-010 chapter titled "Material Control Program" (see 3.12 and 6.6). If the Navy determines that off-gas testing is required, testing shall be conducted at a test facility approved by the Technical Authority (see 3.12). The Navy will review the off-gas test results and assign a usage category. Additionally, the Navy will assign a usage category if an administrative review is conducted in lieu of off-gas testing (see 3.12 and 6.6).

4.4.11 Fire gas toxicity. During the conduct of the smoke density test specified in 3.13, the barrier material shall be evaluated for fire gas toxicity in accordance with ASTM E800. The average value of the maximum value of the gas concentration measured at each condition (flaming and non-flaming) shall be recorded for the following gases, as specified (see 6.2): carbon monoxide (CO), hydrogen chloride (HCl), hydrogen fluoride (HF), nitrogen oxides (NOx), hydrogen bromide (HBr), hydrogen cyanide (HCN), and sulfur dioxide (SO_2). When making toxicity measurements, the sampling of gases shall be made during the testing of the second and the third specimen at each test condition, from the geometrical center of the chamber at the time when the maximum specific optical density of smoke is reached. The number of samples shall be four; two each in the flaming and non-flaming condition. The concentration of each toxic gas shall be determined as parts per million (ppm) in the chamber volume.

4.5 Examinations.

4.5.1 Workmanship. The barrier material shall be visually inspected for the presence of defects listed in table XIX (see 3.11).

4.5.2 Material and visual characteristics. The barrier material shall be visually examined to verify compliance with the material and visual requirements of this specification, including quantity of pieces per roll for type I and type II barrier material (see 3.3.2), and any other requirements specified herein not involving tests.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Barrier materials furnished under this specification are intended for application to plane and curved ship structures and machinery surfaces in order to attenuate airborne noise. Type I and type II barrier materials (see 1.2) are primarily used as the top or middle layer of thermal, acoustic, or fire insulation materials that are attached to the deck, bulkhead, overhead, ducting, or machinery casing surfaces. Standard application details are contained in NAVSEA Drawings 804-5773931 and 804-5773932. The type III barrier material (see 1.2) is intended as a middle layer (sandwich) component of composite treatments and uses an insulation material for the lower layer and a faced panel or metal sheathing as the upper layer. Type III barrier material is used upon Coast Guard approval. Type I, type II, and type III barrier materials are specified in sizes that are convenient and economical for shipboard installation. The barrier materials are cut and fit to irregular shapes, and the sizes are selected to minimize waste and allow handling by installation personnel. Equipment manufactured in accordance with this document may utilize new technology provided form, fit, and function are maintained as approved by the Technical Authority.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type and class required (see 1.2).

c. When first article inspection is required (see 3.1 and 4.2).

d. Requirements for an HHA (see 3.2.1.1 and 4.4.1.1).

e. Requirement for prohibited materials verification (see 3.2.1.2 and 4.4.1.2).

f. Woven outer facing layer location for type II barrier material (see 3.2.3).

g. Width of type I and type II rolls (see 3.3.1).

h. Size of type III panels (see 3.3.3).

i. Requirements for fire gas toxicity (see 3.13, 4.4.11, and 6.8).

j. Inspection conditions, if other than specified (see 4.1.1).

k. When fire resistance, sound transmission loss, and toxicity testing is required for first article inspection (see 4.2.1).

l. Additional fire testing provisions (see 4.4.8.1 and 6.7).

m. Packaging requirements (see 5.1).

n. Requirement for a Safety Data Sheet (SDS) (see 6.4).

6.3 Fabrication. There are many material combinations that potentially could meet the requirements of this document. Three possible material combinations are as follows: (1) Type I barrier material may consist of barium sulfate-loaded vinyl with fibrous glass cloth facing, (2) Type II barrier material may consist of silicone, and (3) Type III barrier material may consist of wire screen mesh hot-roll-bonded to lead sheathing. However, if barrier materials are fabricated from these components, there is no assurance that they will meet all the requirements of this document.

6.4 SDSs. Contracting officers will identify those activities requiring copies of completed SDSs prepared in accordance with FED-STD-313. In order to obtain the SDS, FAR clause 52.223-3 must be in the contract (see 6.2).

6.5 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-Marine-Corps-Public-Health-Center/.

6.6 Material certification. Materials to be installed in submarines are to be controlled to prevent off-gassing, which contaminates the submarine's atmosphere and can result in health hazards to personnel or deleterious effects on machinery. These controls are administered through the Submarine Material Control Program, which is described in the Nuclear Powered Submarine Atmosphere Control Manual, S9510-AB-ATM-010 chapter titled "Material Control Program." Under the Submarine Material Control Program, all materials considered for use on submarines require certification and assignment of a usage category. Under the certification process, candidate materials are selected by Navy activities or contractors, and a request for certification is submitted to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil. The certification request is accompanied by detailed information, including descriptions of the material, method of application, usage, and storage. A chemical analysis is conducted, which can be accomplished through off-gas testing. If off-gas testing is required, it must be conducted in a Government approved laboratory. Information pertaining to this test requirement may be obtained from Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to usn.ncr.comnavseasyscomdc.mbx.command-standards@us.navy.mil. Based on the chemical analysis results, a usage category is assigned to the material defining whether, and to what extent, the material may be used on submarines.

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6.7 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. A2LA (www.A2LA.org) or International Code Council's IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.8 Fire gas toxicity evaluation. The Technical Authority recommends inclusion of the fire gas toxicity evaluation (see 3.13 and 4.4.11). Measurement of the concentration of each toxic gas should be as prescribed in 4.4.11 and provided in the first article test report (see 6.2). The Navy will use the data provided for evaluation purposes.

6.9 Nonconformance. If a sample fails to pass group B conformance inspection, the contractor should notify the approval activity and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product should be discontinued until corrective action acceptable to the contracting activity has been taken. After the corrective action has been taken, group B inspection should be repeated on additional sample units (all tests and examinations, or the test which the original sample failed at the option of the approval activity). Group A inspections may be reinstituted; however, final acceptance and shipment will be withheld until group B inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure should be furnished to the cognizant inspection activity and the contracting activity.

6.10 Cross-reference of classifications and substitutability data. Type II noncombustible barrier material has been redesignated as type III in revision B of this specification. As such, type III noncombustible barrier material in accordance with MIL-PRF-24699B is substitutable for type II noncombustible barrier material in accordance with MIL-PRF-24699A.

6.11 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.12 Subject term (key word) listing.

Breaking strength
Flex stiffness
Peel strength
Surface density
Tearing strength
Warp

6.13 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 5640-2023-015)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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